

Carbon Efficiency for Industrial Infrastructure

A Scalable, Industrial Decarbonisation Infrastructure Platform

For Immediate Scope 1 Reduction



Reduce Fuel. Reduce CO₂. Prove It.

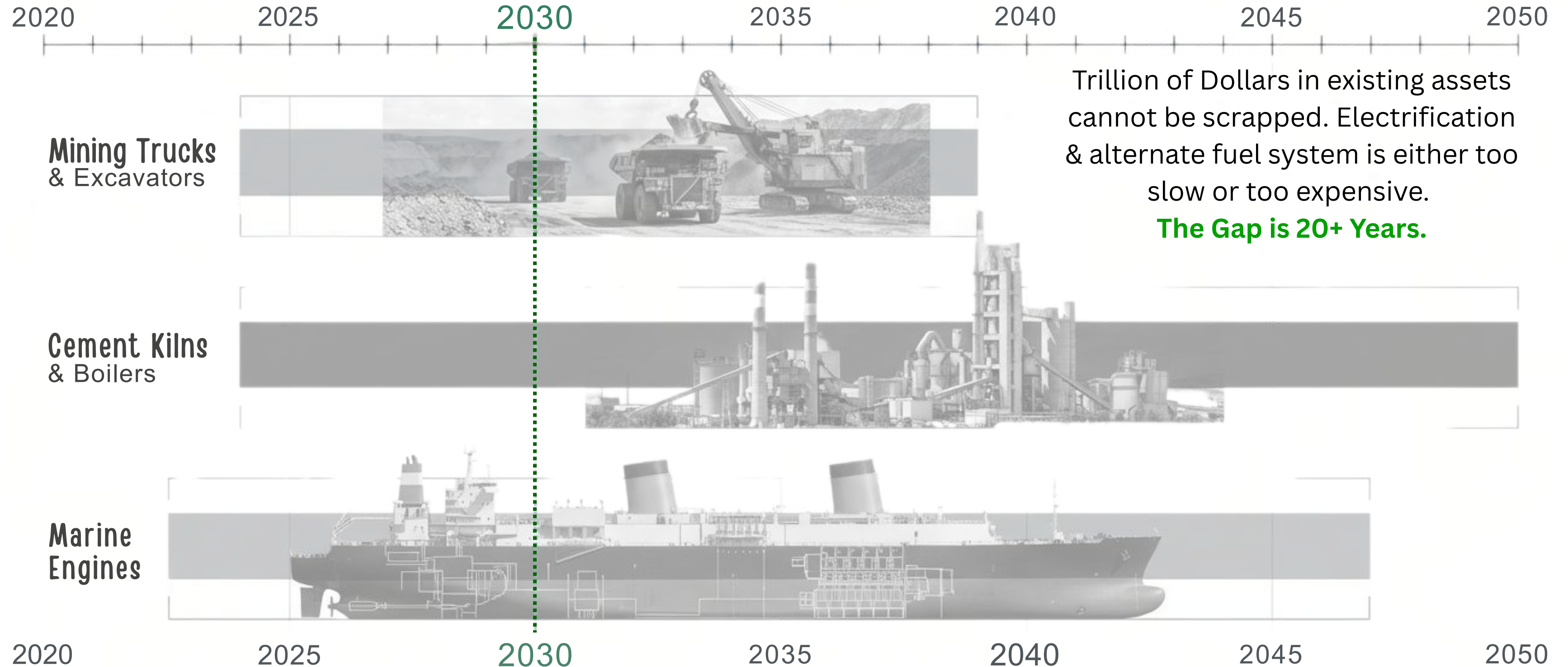
www.sgthydroedge.com



Fossil Fuel Assets Will Operate for Decades



Emission Targets



Direct Combustion Emissions Are Under Scrutiny



Mandatory ESG Reporting

ISSB, EU CSRD, and SEC climate rules are expanding scope.



Global Carbon Pricing

Carbon taxes are spreading to new geographies and sectors.



Capital Mandates

Lenders and investors are restricting capital for high-intensity assets.



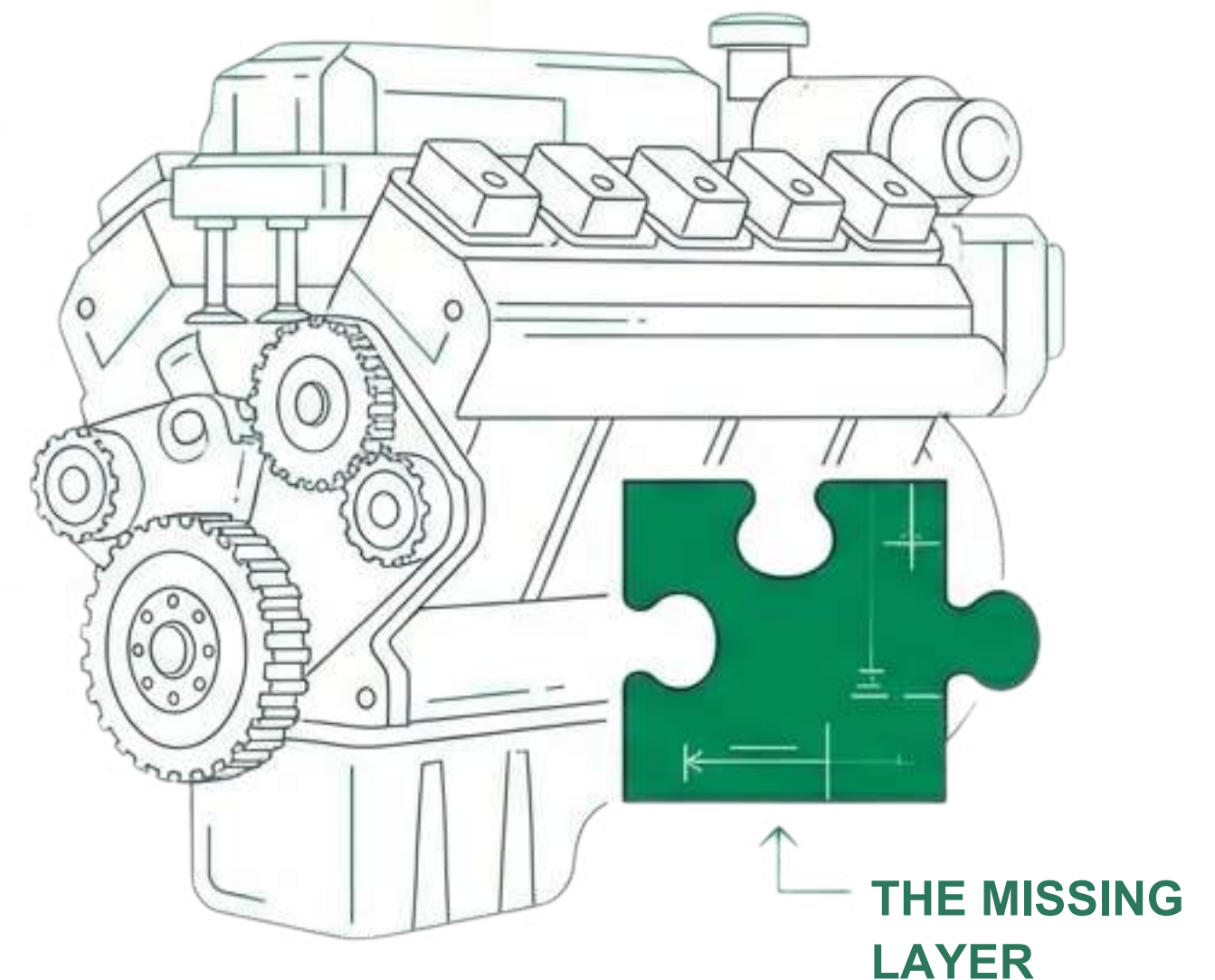
Scope 3 Pressure

Large corporations are forcing vendors to cut their Scope 1 emissions.

Every litre of diesel burned is now accountable.

The Missing Layer In Decarbonisation

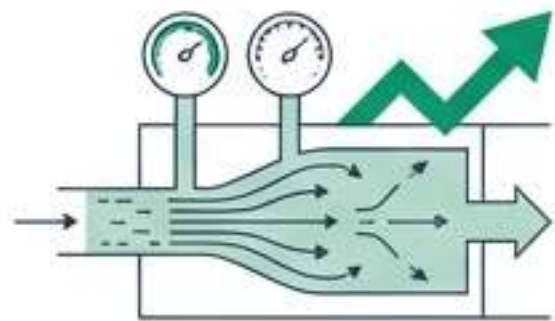
Solution	Pros	Cons
Electrification	Zero Emissions	Low density, lack of grid infra, not ready for heavy industry.
Carbon Capture	Effective	Extremely high CapEx, large physical footprint.
After-Treatment	Reduces pollution	Does not save fuel, Increases maintenance.
SGT HydroEdge	Increases Combustion Efficiency, Immediate ROI, Retrofit	



We Are A Decarbonisation Infrastructure & Solutions Company



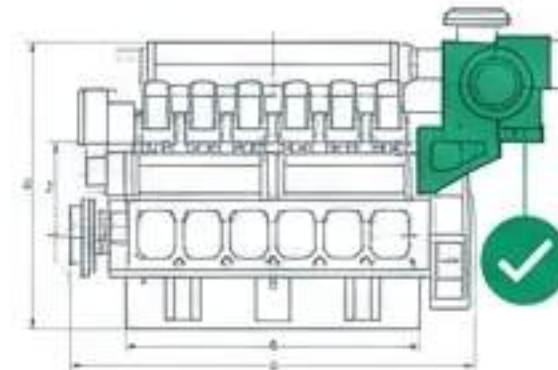
“We optimise combustion at the physics level, validate outcomes with data, and guide decarbonisation through expert consulting.”



Reduce Fuel Consumption
Achieve significant fuel savings through combustion efficiency.



Reduce Emissions
(CO₂, PM, NO_x)
Minimize harmful greenhouse gases and pollutants with immediate impact.



No Engine Replacement
Retrofit existing heavy machinery without capital-intensive engine swaps.

Logistics | Mining | Industrial Manufacturing (Boilers & Kilns) | Marine | DG Power



What Customer Actually Gets

Across Diesel Vehicles, DG Sets, Marine Engines, Boilers, Kilns and every Fossil Fuel asset

3-18% Fuel Reduction



Direct CO₂ Reduction



50-95% PM Reduction, CO & HC by 5-%Plus



Verified ROI & Carbon Dashboards



- NO Changing Fuel



- NO Changing Engines



- NO Storing Hydrogen



- NO Operational Disruption

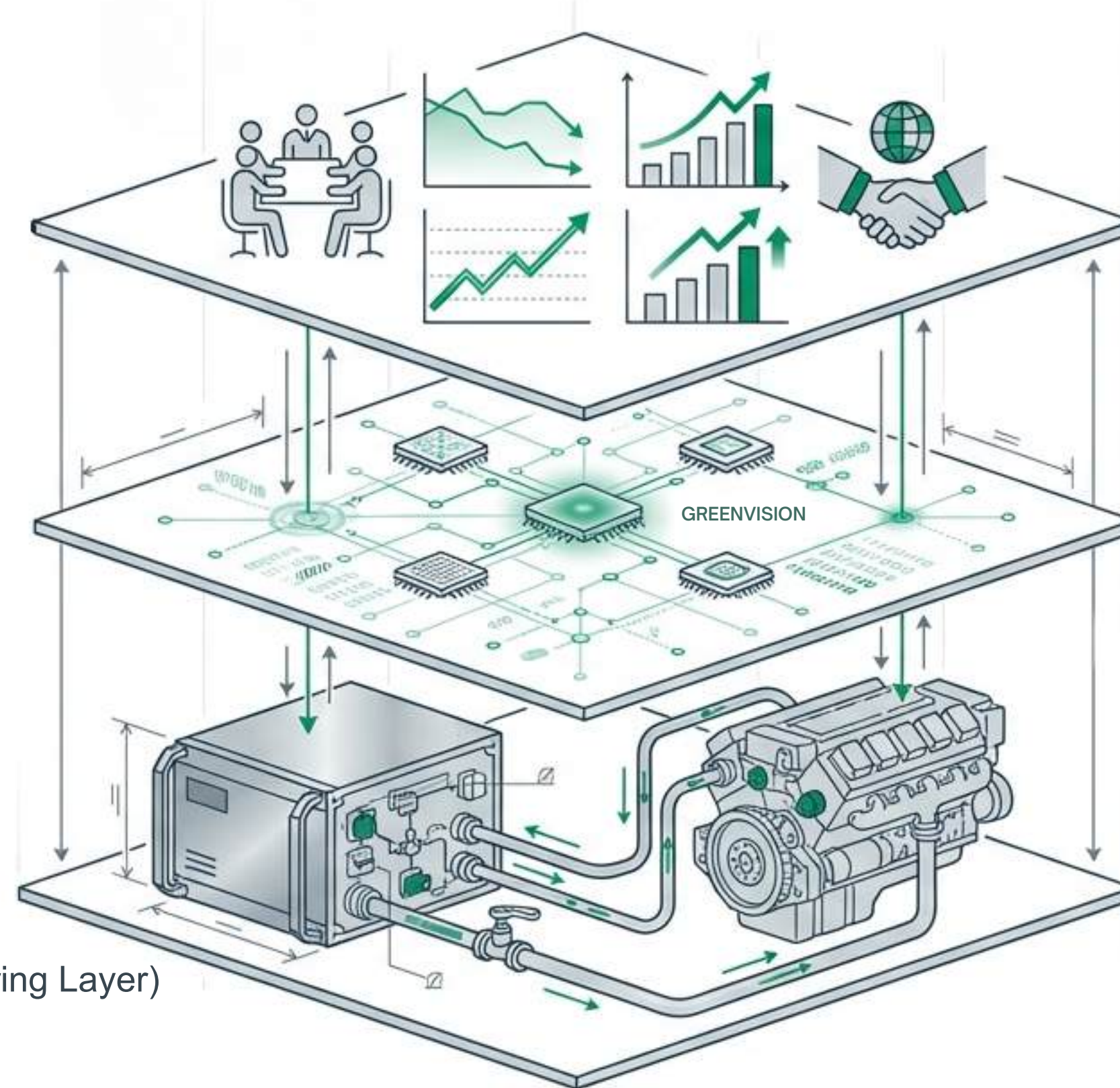
The 3-Layer Integrated Solutions Architecture



LAYER 3:
DECARBONISATION
SERVICES

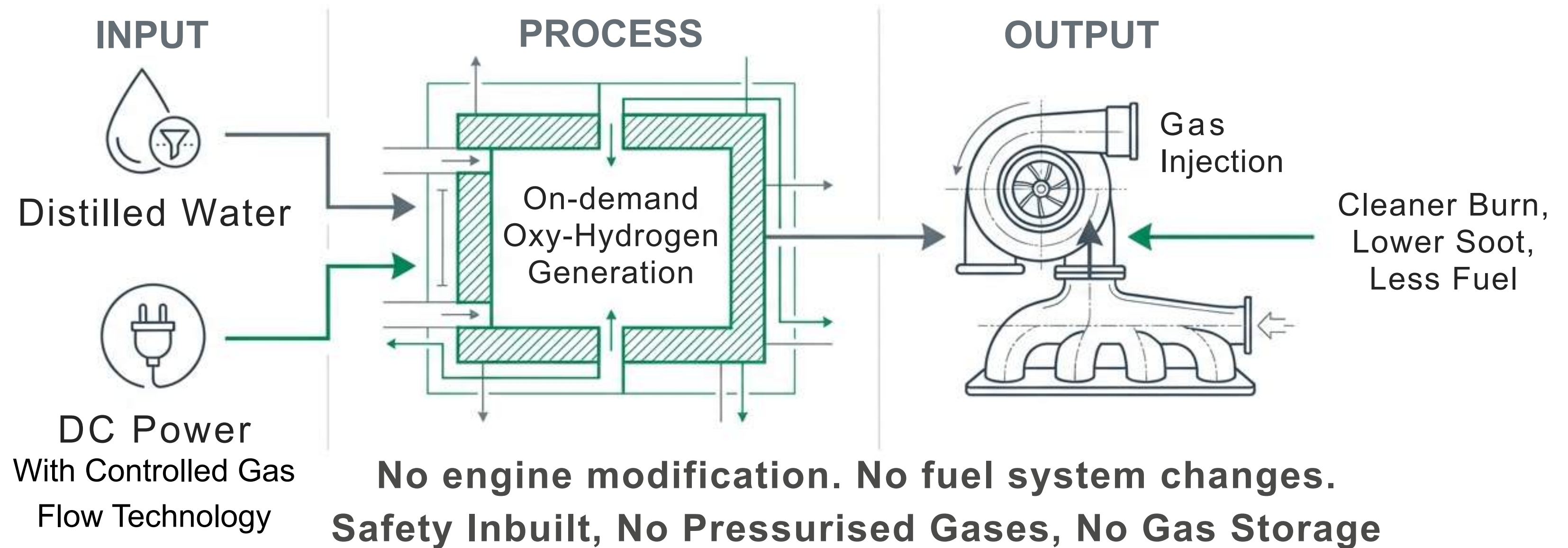
LAYER 2:
DATA &
INTELLIGENCE
(GREENVISION - Our
Data platform)

LAYER 1:
COMBUSTION
OPTIMIZATION
(Our R&D & Manufacturing Layer)



Through Patented
**Controlled Hydrogen Fuel
Assist Technology** with
hardware systems starting
from small vehicle engines
to very large Industrial &
Marine systems

LAYER 1: Controlled Hydrogen Fuel Assist Technology (CHFA)



Scalable, Proven Platform For Every Asset Class



One Core technology tailored for world's most demanding operations.



GreenDrive -For On-Road Vehicles

Delivers 3-20% fuel savings and direct CO₂ reduction in variable load cycles.



GreenX - For Diesel Generators

Provides true fuel & CO₂ reduction and >50% PM cut, even at low loads where other systems fail.



GreenMarine -For Marine Engines

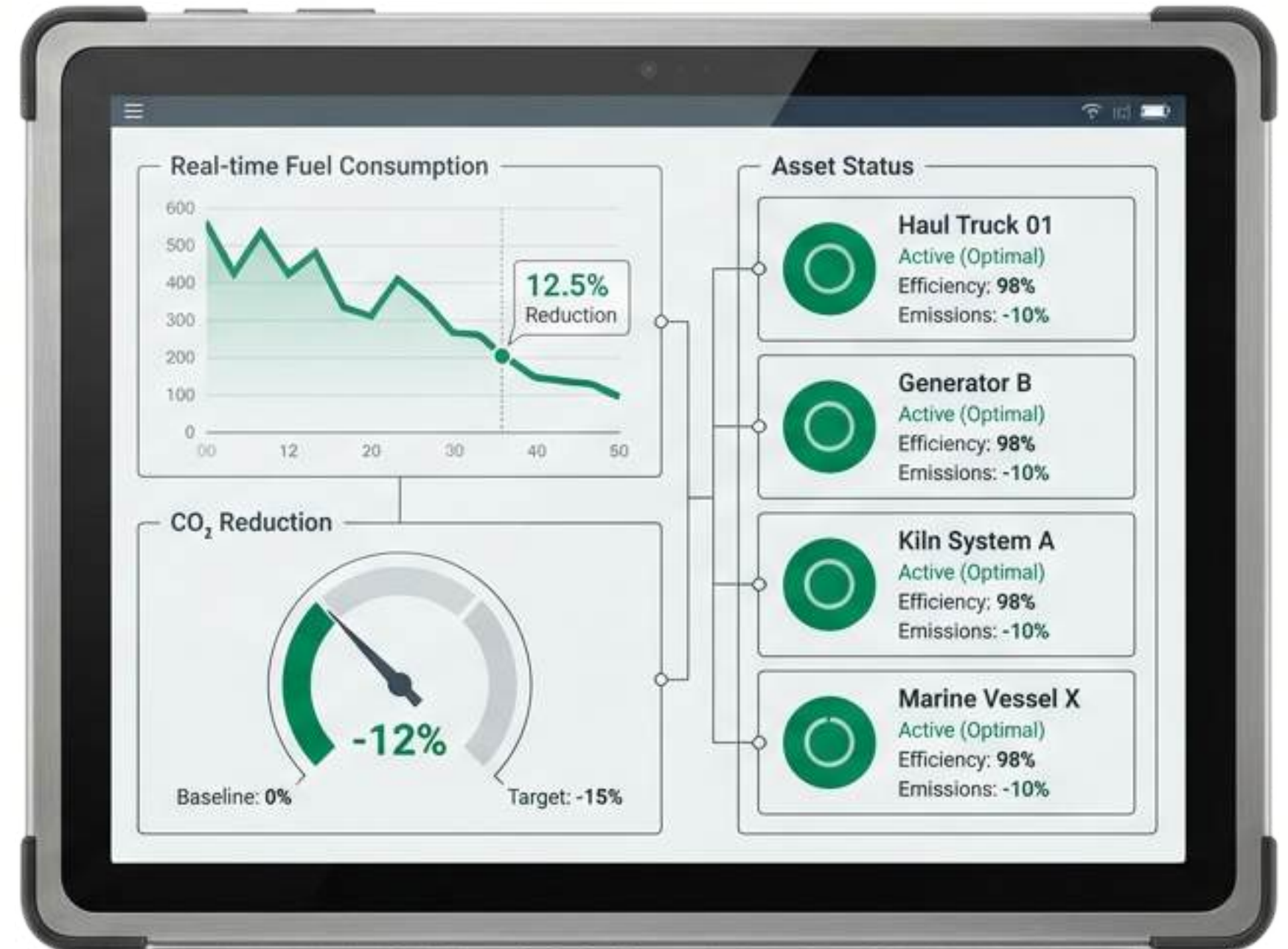
Improves fuel efficiency and reduces emissions without operational disruption.

Industrial HHO Generators Under Development - To be sent for trials by April 2026

Layer 2: GreenVision (Data & Intelligence)

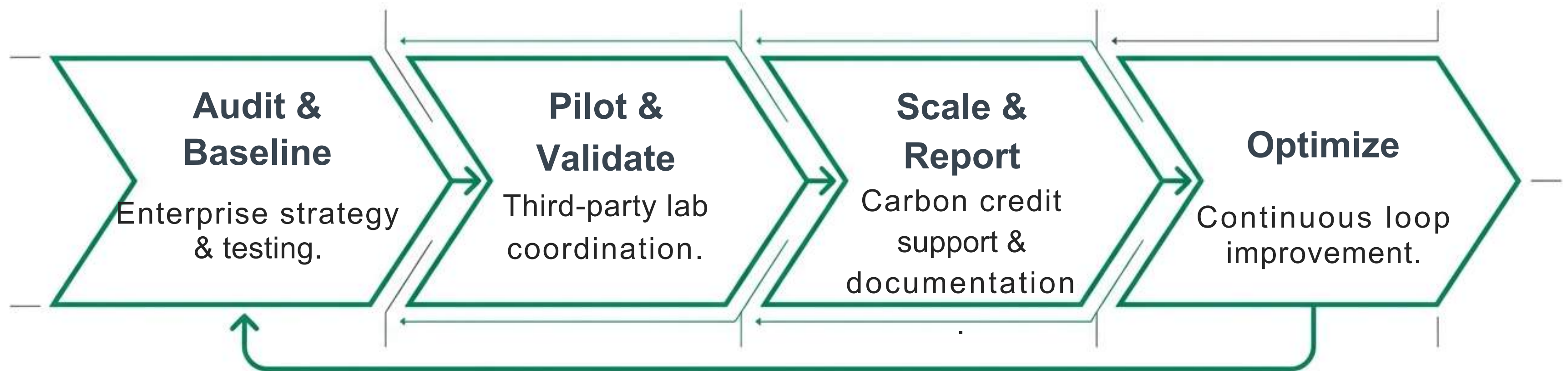
We don't ask customers to trust us, We show them data.

- IoT-based real-time tracking
- Baseline normalization
- ESG-ready reporting



Layer 3: Decarbonisation Services

SGT operates as a long-term partner, not a transactional vendor.



Engineering Depth & Recognitions

Deep combustion engineering. Protected IP. Institutionally validated. Scale-ready.

Global & Institutional Recognition

- Xynteo Energy Leap innovation grant recipient 2025
- CII Ascend POC grant awardee 2026
- Recognised amongst Top Innovation-linked Cleantech company (India) on several forums
- Presented at Jakarta Green Hydrogen Summit
- Technical presence – EGPES 2025 | Finalist – EGPES 2026

DeepTech Combustion Platform

- 3+ years R&D across Logistics, DG, marine & industrial systems
- Indigenous CHFA architecture, engineered & commercialised (Jan 2026)
- Scalable from small engines to large ships, boilers & kilns
- Recognised as DeepTech under Startup India

IP & Scientific Validation

- Multiple patents filed on controlled hydrogen dosing
- Published research with ARAI on HHO based combustion optimisation
- Ongoing ARAI validation for DG emission pathway under CPCB
- Active OEM engagement for mainstream adoption
- More than 50 successful Field trials before launch

Manufacturing & Compliance

- Highly trained and hands on core Team with strong & proven leadership.
- ISO 9001 certified systems & Factory
- ROHS, CE, EMC, FC and all compliances for high grade electronics compliant products
- Designed & manufactured in India for global markets

Strategic Verticals We Serve

Targeting high intensity fuel users for rapid & impactful decarbonisation



**DG POWER
GENERATION**
25 KVA to 4000 KVA



**INDUSTRIAL
MANUFACTURING**
(Boilers, Furnace & Kilns)



MINING



LOGISTICS
Fleet Operators



MARINE
(Auxiliary & Propulsion)



The Fossil Fuel Era Is Not Over. But Inefficiency Is.

The diesel you own. Optimised.

SGT HydroEdge:

Industrial
Decarbonisation
Infrastructure.

*Reduce Fuel.
Reduce CO2.
Prove It.*

Alok Kumar

Founder & Director
alok@sgthydroedge.com
+91 9545451800

www.sgthydroedge.com